Homework 1
STAT 132

1. Suppose that the prior distribution of some parameter θ is a gamma distribution for which the mean is 5 and the variance is 2. Determine the prior pdf of θ.

2. Suppose that the prior distribution of some parameter θ is a beta distribution for which the mean is $2/3$ and the variance is $1/48$. Determine the prior pdf of θ.

3. Suppose you flip a possibly weighted coin 100 times and observe 60 heads. Write the likelihood function for p, the true proportion of heads resulting from this coin. Then, sketch this likelihood function over the $[0, 1]$ interval.

4. Suppose that the number of people living in a household can be modeled using a Poisson distribution with mean λ. You take a random sample of 10 households and observe household sizes $x_1, x_2, \ldots, x_{10}$. Write the likelihood function for λ.

5. Suppose that the number of people living in a household can be modeled using a Poisson distribution with mean λ. You take a random sample of 10 households and observe household sizes $x_1, x_2, \ldots, x_{10}$. Your friend suggests a normal prior distribution with mean 1 and variance 10 for λ. Is this a reasonable prior distribution for this problem? Why or why not?